Kathaa Senapati

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EDUCATION

Università degli studi di Padova **Oct 2023 - Jun 2026**
Master's in Astrophysics and Cosmology *Padova, Italy*
Final Grade: GPA - 110/110

Collegio Universitario Don Nicola Mazza **Sept 2023 - July 2026**
Diploma di Collegio di Merito *Padova, Italy*

Bennett University **July 2019 - May 2023**
Bachelor's of Technology in Engineering Physics *Uttar Pradesh, India*
Final Grade: GPA - 8.6/10

EXPERIENCE

Università degli studi di Padova **Apr 2025 - Jun 2026**
Master's Thesis Project *Padova, Italy*

- Investigation of the RGB bump in globular clusters using Hubble Space Telescope photometric data
- Stellar Astrophysics, Photometry, RGB Luminosity Function Modelling, Python Data Analysis

Spartificial **Aug 2023 - Oct 2023**
Machine Learning Research Intern *Rajasthan, India*

- Prediction of asteroid diameters using astronomical observational data
- Python, Machine learning

University of Calcutta **Jan 2023 - May 2023**
Research Internship - Bachelor's Thesis Project *West Bengal, India*

- Mathematical modelling of prey-predator dynamics using Holling Type III functional response and Allee effect
- Nonlinear Dynamics, Stability and Bifurcation Analysis, MATLAB, Python

Bennett University **Aug 2022 - Dec 2022**
Academic Major Project *Uttar Pradesh, India*

- Calculation of surface tension to characterize Quantum Liquid in Quasi One-Dimensional Bose-Einstein Condensate
- Mathematical Modelling, Analytical Calculations, Wolfram Mathematica

Indian Institute of Technology (IIT) Indore **Jun 2022 - Aug 2022**
Research Internship (Remote) *Madhya Pradesh, India*

- Analysis of plane-wave propagation in isotropic thermoelastic solids

Hindalco Industries Ltd. **Jun 2022 - Jul 2022**
Summer Internship *West Bengal, India*

- Analysis of power consumption, heat loss and efficiency of an aluminium annealing furnace
- Data Analysis, MS Excel

RELEVANT SKILLS

Programming Languages: Python (AstroPy, scikit-learn, NumPy, Matplotlib), C/C++, SQL
Software and Tools: VS Code, Linux, GitHub, Docker, MATLAB, Wolfram Mathematica, LaTex, Microsoft Office (Word, Excel, Powerpoint)
Technical Skills: Mathematical Modelling, Data Analysis, Research analysis
Soft Skills: Teamwork & collaboration, Critical Thinking, Problem solving, Adaptability

PUBLICATIONS

Consequences of Allee effect on the multiple limit cycles in a predator-prey model

- Mondal, B., Senapati, K., Pandey, S., & Ghosh, U. (2023). Consequences of Allee Effect on the Multiple Limit Cycles in a Predator–Prey Model. The European Physical Journal Plus, 138, Article 919

Quantum Liquid in Lower Dimensions: From the Perspective of Surface Tension

- Adusumalli, S. S., Senapati, K., Singh, S., & Khan, A.(2024). Quantum Liquid in Lower Dimensions: From the Perspective of Surface Tension. Physics Letters A, 516, 129638

EXTRACURRICULAR

Collegio Universitario Don Nicola Mazza, Padova, Italy

- International Committee Member, 2023 - 2026
- ArteLab Committee Member, 2023 - 2026
- International Dinner Event Organizer and Participant, 2025
- International Mobility Program: Travel with Awareness, Italy and Austria , 2025

Bennett University, Uttar Pradesh, India

- International Symposium on Advances in Nano sensors & Nanomedicine, 2021
- Mathematica Basic Training with Hands-on session, 2020

Indian Institute of Astrophysics (IIA), Karnataka, India

- Summer School Programme, 2021

CERTIFICATIONS

- Training program on Astronomical Data Science with Python, Spartificial, 2023
- Robotics Course, Skyfi Labs, 2020

LANGUAGES

- English: C1
- Italian: A1
- Bengali: Native
- Hindi: C2